Thrinadh Sai Nimmagadda

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building backend systems and APIs using Python, Java, and SQL, with hands-on experience in NLP, RAG, and LLM applications. Improved parsing accuracy from 50% to 80% and reduced API latency by 20%, with experience across distributed systems, Kafka, Docker, and AWS.

WORK EXPERIENCE

Artificial Intelligence Institute of South Carolina | AI Engineer Intern May 2025 - Aug 2025

- Built a full-stack research intelligence platform supporting semantic discovery, topic clustering, single-paper RAG, and citation-grounded multi-document question answering across 100 academic papers.
- Engineered a hybrid retrieval pipeline combining BM25 and dense embeddings, retrieving 20 candidate passages and applying cross-encoder reranking to select the five most relevant passages for LLM context.
- Reduced end-to-end query latency from 41.2s on the initial cold path to 18.4s on the optimized warm path through model caching and bounded local Llama generation.
- Built a retrieval and performance evaluation harness measuring Recall@K, MRR, mean latency, and p95 latency across representative queries.

Wipro | Software Engineer Jul 2021 - Aug 2024

Client - Meta ads

- Contributed to Java/Spring Boot microservices supporting Meta Ads event-processing workflows; implemented REST API changes, SQL persistence, data mappings, and validation logic for advertiser data.
- Monitored and optimized Apache Kafka event streams across microservices; analyzed payloads and logs to resolve message processing failures and ensure downstream data consistency.
- Improved advertiser event data quality through field validation, normalization, and duplicate-event handling, reducing malformed and incomplete event-processing failures by ~30%.
- Investigated API latency and production issues using service logs, Postman, and database tracing; optimized SQL operations and reduced unnecessary database round trips, improving average response time by ~20%, while adding regression tests for recurring issues.

PROJECTS

Incident Copilot

- Built an AI-driven incident diagnosis pipeline using Python and Docker to automate enterprise log analysis at scale.
- Implemented Kafka-based real-time streaming and anomaly detection with design patterns focused on system reliability and scaling.
- Delivered a FastAPI web dashboard with live diagnosis and approval workflows through iterative development emphasizing software excellence.

ClinicalMatch AI

- Built an end-to-end clinical trial matching system leveraging SQL and Python for data ingestion and retrieval.
- Designed and deployed the application using Docker and FastAPI, owning the system architecture and major deliverables end-to-end.
- Reduced LLM operational costs by 13x through optimized retrieval with LangChain and ChromaDB automation at scale.

CogniTeam

- Constructed ML pipelines for predicting burnout, attrition, and conflict risk using XGBoost and SHAP explainability.

- Instituted GenAI guardrails to safeguard privacy and prevent misuse, including blocking private-message disclosures and punitive requests.
- Deployed a privacy-preserving backend architecture featuring differential privacy, audit logging, and SHA-256 anonymization.

SKILLS

Programming Languages: Java, Python, SQL, C++

Cloud and Database: AWS, RDS, PostgreSQL, MySQL

Machine Learning & AI: LLMs, RAG Pipelines, NLP, XGBoost, scikit-learn, TensorFlow, PyTorch, SHAP

Frameworks & Libraries: FastAPI, Django, Flask, Spring Boot, React, LangChain, LangGraph, Hugging Face, Sentence- Transformers.

Databases & Tools: PostgreSQL, MySQL, Redis, Docker, Git, CI/CD, Kafka, ChromaDB, Power BI.

CERTIFICATIONS & ACHIEVEMENTS

- Azure Fundamentals - Microsoft — Nov 2023

EDUCATION

University of South Carolina — Aug 2024 - Dec 2026
MS, *Computer Science* — GPA: 3.5

B.V. Raju Institute of Technology — Jul 2018 - Jun 2022
Bachelor's, *Computer Science* — GPA: 4.0